

Brutal Practice Paper B54 - Acids Bases & Salts, Fractions & Decimals, Reproduction in Plants, Simple Equations

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A farmer has 600 kg of lime. He tested 8 fields with blue litmus paper; three-quarters of them turned the paper red. Each acidic field needs 75 kg of lime. How much lime is left after treating every acidic field? [[Acids, Bases & Salts](#)]

- (A) 225 kg
- (B) 300 kg
- (C) 450 kg
- (D) 525 kg
- (E) 0 kg
- (F) 150 kg
- (G) 100 kg

2. A bottle holds $\frac{3}{4}$ litre of juice. A glass takes $\frac{1}{8}$ litre. After filling 4 glasses from the bottle, how much juice is left? [[Fractions & Decimals](#)]

- (A) 0 litre
- (B) $\frac{3}{8}$ litre
- (C) $\frac{5}{8}$ litre
- (D) $\frac{1}{4}$ litre
- (E) $\frac{3}{4}$ litre
- (F) $\frac{1}{8}$ litre

3. One yeast cell reproduces by budding and doubles every 20 minutes. Starting from a single cell, how many yeast cells are there after 2 hours? [[Reproduction in Plants](#)]

- (A) 24
- (B) 32
- (C) 36
- (D) 6
- (E) 128
- (F) 12
- (G) 64

4. Think of a number. Multiply it by 4, then subtract 7, and the result is 25. What is the number? [[Simple Equations](#)]

- (A) 9
- (B) 4.5
- (C) 18
- (D) 7
- (E) 8
- (F) 12
- (G) 5

5. A bee sting injects an acid under the skin. A nurse uses a mild base to cancel it, where one spoon of base neutralises 4 mL of acid. A sting left 18 mL of acid. How many FULL spoons are needed to neutralise all of it? [\[Acids, Bases & Salts\]](#)

- (A) 9 spoons
- (B) 2 spoons
- (C) 3 spoons
- (D) 4 spoons
- (E) 6 spoons
- (F) 5 spoons

6. A recipe needs 0.35 kg of sugar for one cake. You bake 6 cakes from a jar holding 2.5 kg of sugar. How much sugar is left? [\[Fractions & Decimals\]](#)

- (A) 0.4 kg
- (B) 0.6 kg
- (C) 0.1 kg
- (D) 0.35 kg
- (E) 0.45 kg
- (F) 2.1 kg
- (G) 0.5 kg

7. A flower has 5 stamens, and each anther holds 800 pollen grains. If only 15% of all the pollen lands on a stigma, how many grains land? [\[Reproduction in Plants\]](#)

- (A) 3400
- (B) 4000
- (C) 400
- (D) 600
- (E) 60
- (F) 1200

8. When 5 is added to twice a number, the result is 23. Find the number. [\[Simple Equations\]](#)

- (A) 13
- (B) 28
- (C) 18
- (D) 4
- (E) 9
- (F) 14
- (G) 11

9. Turmeric paper stays yellow in acids and turns red in bases. A student tests lemon juice, soap solution, vinegar, lime water and sugar water. How many of these turn the turmeric paper red? [\[Acids, Bases & Salts\]](#)

- (A) 4
- (B) 6
- (C) 5
- (D) 0
- (E) 1
- (F) 2

10. Which of these decimals is the largest: 0.7, 0.07, 0.77, 0.707, 0.077? [\[Fractions & Decimals\]](#)

- (A) 0.07
- (B) 0.077
- (C) 0.77
- (D) 0.7
- (E) 0.707
- (F) 0.7 and 0.77 are equal
- (G) they are all equal

11. A farmer sows 500 seeds, and 8 out of every 10 germinate. How many seeds fail to germinate?

[\[Reproduction in Plants\]](#)

- (A) 100
- (B) 250
- (C) 80
- (D) 20
- (E) 50
- (F) 125
- (G) 400

12. One-third of a number is 12. What is the number? [\[Simple Equations\]](#)

- (A) 9
- (B) 48
- (C) 15
- (D) 4
- (E) 24
- (F) 39
- (G) 36

13. An acid is neutralised by a base in the volume ratio 2 parts acid to 3 parts base. A tank holds 40 litres of the acid. How many litres of base are needed to neutralise it exactly? [\[Acids, Bases & Salts\]](#)

- (A) 120 litres
- (B) 27 litres
- (C) 100 litres
- (D) 80 litres
- (E) 24 litres
- (F) 26.7 litres
- (G) 60 litres

14. A car uses 0.08 litre of petrol per kilometre. How many kilometres can it travel on 6 litres of petrol?

[\[Fractions & Decimals\]](#)

- (A) 48 km
- (B) 60 km
- (C) 0.75 km
- (D) 80 km
- (E) 750 km
- (F) 75 km
- (G) 0.48 km

15. A fern's spore-case bursts and releases 2000 spores, but only 1 in every 250 grows into a new plant. How many new plants form? [\[Reproduction in Plants\]](#)

- (A) 500
- (B) 16
- (C) 250
- (D) 8
- (E) 80
- (F) 4

16. A number is multiplied by 6 and then 8 is added, giving 50. Find the number. [\[Simple Equations\]](#)

- (A) 7
- (B) 9.67
- (C) 42
- (D) 9
- (E) 8
- (F) 6

17. Excess stomach acid causes pain. One antacid tablet neutralises 5 mL of acid. The stomach makes 2 mL of extra acid every 10 minutes. For how many minutes does one tablet keep the extra acid neutralised?

[\[Acids, Bases & Salts\]](#)

- (A) 20 minutes
- (B) 12.5 minutes
- (C) 25 minutes
- (D) 5 minutes
- (E) 15 minutes
- (F) 50 minutes

18. In a school, $\frac{2}{3}$ of the students play a sport, and of those, $\frac{3}{4}$ play cricket. If there are 240 students, how many play cricket? [\[Fractions & Decimals\]](#)

- (A) 90
- (B) 160
- (C) 60
- (D) 80
- (E) 120
- (F) 180
- (G) 200

19. A potato has 12 eyes (buds), and three-quarters of them sprout into new plants. From 20 such potatoes, how many new plants grow? [\[Reproduction in Plants\]](#)

- (A) 9
- (B) 36
- (C) 135
- (D) 144
- (E) 60
- (F) 180
- (G) 240

20. The sum of three consecutive whole numbers is 72. Find the smallest of the three. [\[Simple Equations\]](#)

- (A) 25
- (B) 24
- (C) 36
- (D) 22
- (E) 21
- (F) 69
- (G) 23

21. An ant bite leaves formic acid; rubbing baking soda neutralises it, and each bite needs 2.5 g of soda. A first-aid box has 30 g of baking soda. If 8 children each get bitten twice, is there enough soda? [\[Acids, Bases & Salts\]](#)

- (A) Yes, exactly enough
- (B) No, short by 25 g
- (C) No, short by 20 g
- (D) No, short by 10 g
- (E) No, short by 5 g
- (F) Yes, 14 g spare
- (G) Yes, 10 g spare

22. A water tank is $\frac{4}{5}$ full. After 30 litres are used, it is exactly $\frac{1}{2}$ full. What is the tank's full capacity? [\[Fractions & Decimals\]](#)

- (A) 100 litres
- (B) 120 litres
- (C) 150 litres
- (D) 75 litres
- (E) 50 litres
- (F) 80 litres

23. A flower has 6 stamens, and each anther has 2 pollen-holding lobes. How many anther lobes does the flower have in total? [\[Reproduction in Plants\]](#)

- (A) 2
- (B) 12
- (C) 8
- (D) 7
- (E) 14
- (F) 24

24. A pencil costs Rs x . Buying 5 pencils and one Rs 10 eraser costs Rs 45 in total. Find the cost of one pencil. [\[Simple Equations\]](#)

- (A) Rs 6
- (B) Rs 9
- (C) Rs 35
- (D) Rs 7
- (E) Rs 5
- (F) Rs 8
- (G) Rs 11

25. China rose indicator turns deep pink in acids and green in bases. Out of 6 samples, 4 turned the solution green. What fraction of the samples were acidic? [\[Acids, Bases & Salts\]](#)

- (A) $\frac{1}{3}$
- (B) $\frac{2}{3}$
- (C) $\frac{4}{6}$
- (D) $\frac{1}{4}$
- (E) $\frac{1}{2}$
- (F) $\frac{5}{6}$

26. Multiply 2.5 by 0.4 and then add 1.5 to the result. What do you get? [\[Fractions & Decimals\]](#)

- (A) 2.5
- (B) 2.9
- (C) 11.5
- (D) 2.0
- (E) 1.0
- (F) 3.5

27. Pollen from a flower lands on the stigma of the SAME flower. What is this process called? [\[Reproduction in Plants\]](#)

- (A) Cross-pollination
- (B) Fertilisation
- (C) Self-pollination
- (D) Budding
- (E) Germination
- (F) Seed dispersal
- (G) Vegetative propagation

28. If 3 times the quantity $(x + 4)$ equals 27, find x . [\[Simple Equations\]](#)

- (A) 13
- (B) 5
- (C) 3
- (D) 7
- (E) 11
- (F) 9
- (G) 1

29. Acid is added drop by drop to a pink basic solution until it just turns colourless. It took 24 drops to neutralise 8 mL of base. How many drops would neutralise 20 mL of the same base? [\[Acids, Bases & Salts\]](#)

- (A) 24 drops
- (B) 80 drops
- (C) 12 drops
- (D) 60 drops
- (E) 72 drops
- (F) 48 drops
- (G) 30 drops

30. A rod 2.4 m long is cut into pieces each 0.3 m long. How many pieces are made? [\[Fractions & Decimals\]](#)

- (A) 80
- (B) 0.8
- (C) 8
- (D) 9
- (E) 6
- (F) 2.1

31. 100 dandelion seeds blow away in the wind. 30% land on fertile soil, and half of those germinate. How many new plants grow? [\[Reproduction in Plants\]](#)

- (A) 45
- (B) 30
- (C) 15
- (D) 50
- (E) 20
- (F) 3
- (G) 60

32. Twice a number minus 9 equals the number plus 4. Find the number. [\[Simple Equations\]](#)

- (A) 13
- (B) 5
- (C) 6.5
- (D) 2.5
- (E) 11
- (F) 26
- (G) -5

33. An acid and a base react to give only a salt and water. 35 g of acid reacts fully with 25 g of base. If 12 g of water forms, what mass of salt is produced? [\[Acids, Bases & Salts\]](#)

- (A) 48 g
- (B) 23 g
- (C) 40 g
- (D) 50 g
- (E) 60 g
- (F) 12 g
- (G) 72 g

34. Find half of three-quarters of 48. [\[Fractions & Decimals\]](#)

- (A) 12
- (B) 36
- (C) 9
- (D) 72
- (E) 18
- (F) 16

35. After pollination, the male cell from the pollen joins the egg inside the ovule. What is this joining called, and what does the ovule become? [\[Reproduction in Plants\]](#)

- (A) Fertilisation; the ovule becomes a fruit
- (B) Fertilisation; the ovule becomes a flower
- (C) Fertilisation; the ovule becomes a seed
- (D) Pollination; the ovule becomes a fruit
- (E) Germination; the ovule becomes a seed
- (F) Dispersal; the ovule becomes a seed
- (G) Budding; the ovule becomes a seed

36. A father is 4 times as old as his son. Together their ages add up to 50 years. How old is the son? [\[Simple Equations\]](#)

- (A) 12.5 years
- (B) 46 years
- (C) 10 years
- (D) 5 years
- (E) 8 years
- (F) 40 years

37. From this list, how many are ACIDS: vinegar, milk of magnesia, curd, soap, tamarind, lime water? [\[Acids, Bases & Salts\]](#)

- (A) 1
- (B) 5
- (C) 6
- (D) 2
- (E) 3
- (F) 0
- (G) 4

38. Which is bigger, $\frac{5}{8}$ or 0.6, and by how much (as a decimal)? [\[Fractions & Decimals\]](#)

- (A) 0.6 is bigger by 0.04
- (B) $\frac{5}{8}$ is bigger by 0.25
- (C) 0.6 is bigger by 0.25
- (D) 0.6 is bigger by 0.025
- (E) $\frac{5}{8}$ is bigger by 0.025
- (F) they are equal
- (G) $\frac{5}{8}$ is bigger by 0.125

39. A pea pod is a fruit formed from a single ovary. If that ovary contained 8 ovules and all were fertilised, how many seeds (peas) are inside the pod? [\[Reproduction in Plants\]](#)

- (A) 1 seed
- (B) 4 seeds
- (C) 16 seeds
- (D) 7 seeds
- (E) 9 seeds
- (F) 8 seeds
- (G) 2 seeds

40. The perimeter of a square is 36 cm. Form an equation and find the length of one side. [\[Simple Equations\]](#)

- (A) 18 cm
- (B) 9 cm
- (C) 4.5 cm
- (D) 6 cm
- (E) 8 cm
- (F) 12 cm

41. A factory releases 5000 litres of acidic water each day. Each litre needs 0.02 kg of base to neutralise. How many kilograms of base are needed per day? [\[Acids, Bases & Salts\]](#)

- (A) 100 kg
- (B) 10 kg
- (C) 2500 kg
- (D) 25 kg
- (E) 200 kg
- (F) 1000 kg
- (G) 50 kg

42. A painter covers $\frac{2}{5}$ of a wall on the first day and $\frac{1}{3}$ of it on the second day. What fraction of the wall is still left? [\[Fractions & Decimals\]](#)

- (A) $\frac{1}{15}$
- (B) $\frac{2}{15}$
- (C) $\frac{4}{15}$
- (D) $\frac{3}{8}$
- (E) $\frac{7}{15}$
- (F) $\frac{8}{15}$

43. A yeast cell doubles by budding every 15 minutes. Starting from one cell, how many cells are there after 1 hour? [\[Reproduction in Plants\]](#)

- (A) 24
- (B) 4
- (C) 32
- (D) 16
- (E) 8
- (F) 12
- (G) 60

44. Five less than three times a number is 19. Find the number. [\[Simple Equations\]](#)

- (A) 6
- (B) 4.67
- (C) 14
- (D) 7
- (E) 8
- (F) 9

45. A lake holding 1,50,000 litres has turned acidic and needs 0.5 g of lime per litre to fix. How many kilograms of lime are needed? [\[Acids, Bases & Salts\]](#)

- (A) 75 kg
- (B) 37.5 kg
- (C) 750 kg
- (D) 300 kg
- (E) 75000 kg
- (F) 150 kg

46. Write 0.375 as a fraction in its lowest terms. [\[Fractions & Decimals\]](#)

- (A) $\frac{3}{4}$
- (B) $\frac{5}{8}$
- (C) $\frac{3}{80}$
- (D) $\frac{3}{8}$
- (E) $\frac{375}{1000}$
- (F) $\frac{1}{3}$
- (G) $\frac{37}{100}$

47. A fern frond has 40 spore-cases, and each spore-case holds 64 spores. How many spores can one frond release in total? [\[Reproduction in Plants\]](#)

- (A) 25600
- (B) 256
- (C) 2500
- (D) 2056
- (E) 640
- (F) 104
- (G) 2560

48. A number is divided by 4, and then 3 is subtracted, giving 5. Find the number. [\[Simple Equations\]](#)

- (A) 23
- (B) 28
- (C) 2
- (D) 32
- (E) 17
- (F) 8
- (G) 20

49. Three solutions A, B and C are tested. A turns blue litmus red, B turns red litmus blue, and C changes neither colour. Which one behaves like common salt dissolved in water? [\[Acids, Bases & Salts\]](#)

- (A) B and C
- (B) C
- (C) B
- (D) all of them
- (E) A and B
- (F) none of them

50. A bolt of cloth is $3\frac{1}{2}$ m long. Each dress needs $\frac{7}{8}$ m of cloth. How many full dresses can be made?

[Fractions & Decimals]

- (A) 7 dresses
- (B) 2 dresses
- (C) 4 dresses
- (D) 28 dresses
- (E) 6 dresses
- (F) 5 dresses

51. An orchard has 250 apple trees that need bees to cross-pollinate. If bees visit only 60% of the trees, how many trees are NOT pollinated by bees? [Reproduction in Plants]

- (A) 40
- (B) 90
- (C) 100
- (D) 60
- (E) 25
- (F) 190

52. Two numbers differ by 8 and add up to 30. Find the larger number. [Simple Equations]

- (A) 23
- (B) 21
- (C) 11
- (D) 22
- (E) 38
- (F) 15
- (G) 19

53. An antacid neutralises 5 mL of acid on an empty stomach but only 4 mL when taken with food. A patient has 90 mL of extra acid and takes the tablets with food. How many tablets are needed? [Acids, Bases & Salts]

[Acids, Bases & Salts]

- (A) 18 tablets
- (B) 19 tablets
- (C) 20 tablets
- (D) 90 tablets
- (E) 23 tablets
- (F) 22 tablets
- (G) 24 tablets

54. Pens cost Rs 12.50 each. With Rs 100, how many pens can you buy and how much money is left?

[Fractions & Decimals]

- (A) 8 pens, Rs 5 left
- (B) 8 pens, Rs 12.50 left
- (C) 8 pens, Rs 10 left
- (D) 6 pens, Rs 25 left
- (E) 8 pens, Rs 0 left
- (F) 9 pens, Rs 0 left

55. A sunflower head has 800 tiny florets, and each pollinated floret forms one seed. If $\frac{3}{8}$ of the florets are NOT pollinated, how many seeds form? [\[Reproduction in Plants\]](#)

- (A) 5
- (B) 800
- (C) 300
- (D) 200
- (E) 100
- (F) 500
- (G) 533

56. When 7 is added to a number and the sum is then doubled, the result is 30. Find the number. [\[Simple Equations\]](#)

- (A) 16
- (B) 11.5
- (C) 23
- (D) 4
- (E) 8
- (F) 2

57. 20 mL of an acid is neutralised by 50 mL of a base. How much of the same base is needed to neutralise 70 mL of the acid? [\[Acids, Bases & Salts\]](#)

- (A) 175 mL
- (B) 350 mL
- (C) 28 mL
- (D) 100 mL
- (E) 120 mL
- (F) 87.5 mL
- (G) 140 mL

58. Add $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$. [\[Fractions & Decimals\]](#)

- (A) 1
- (B) $\frac{9}{16}$
- (C) $\frac{4}{16}$
- (D) $\frac{15}{16}$
- (E) $\frac{13}{16}$
- (F) $\frac{11}{16}$
- (G) $\frac{7}{16}$

59. Seeds need water, air and warmth to germinate. Out of 60 soaked seeds, one-third received no water at all. How many seeds can still germinate? [\[Reproduction in Plants\]](#)

- (A) 30
- (B) 20
- (C) 45
- (D) 60
- (E) 50
- (F) 15
- (G) 40

60. A rope 48 m long is cut into two parts, where one part is 3 times the other. Find the length of the shorter part. [\[Simple Equations\]](#)

- (A) 36 m
- (B) 4 m
- (C) 16 m
- (D) 12 m
- (E) 8 m
- (F) 24 m
- (G) 9 m

61. During a meal the stomach makes 120 mL of acid. An antacid removes 35% of it. How much acid remains? [\[Acids, Bases & Salts\]](#)

- (A) 84 mL
- (B) 85 mL
- (C) 42 mL
- (D) 48 mL
- (E) 78 mL
- (F) 162 mL
- (G) 35 mL

62. If 0.25 of a number is 19, what is the number? [\[Fractions & Decimals\]](#)

- (A) 19.25
- (B) 95
- (C) 38
- (D) 4.75
- (E) 57
- (F) 76

63. A strawberry plant grows new plants along runners; each runner produces 3 new plants. If 4 parent plants each send out 2 runners, how many NEW plants form? [\[Reproduction in Plants\]](#)

- (A) 24
- (B) 6
- (C) 12
- (D) 8
- (E) 30
- (F) 20
- (G) 14

64. If two-thirds of a number equals 10, find the number. [\[Simple Equations\]](#)

- (A) 30
- (B) 7.5
- (C) 20
- (D) 5
- (E) 15
- (F) 13
- (G) 6.67

65. A village has 20 fields: 6 are acidic, 9 are basic, and the rest are neutral. Acidic fields get lime and basic fields get organic manure. How many fields need NO treatment? [\[Acids, Bases & Salts\]](#)

- (A) 5
- (B) 6
- (C) 11
- (D) 15
- (E) 0
- (F) 9

66. A tap fills a bucket at 0.6 litre per second. How many seconds does it take to fill a 27 litre bucket?

[\[Fractions & Decimals\]](#)

- (A) 50 seconds
- (B) 16.2 seconds
- (C) 4.5 seconds
- (D) 45 seconds
- (E) 45.6 seconds
- (F) 27.6 seconds

67. Of 200 flowers, $\frac{1}{2}$ are wind-pollinated and $\frac{1}{4}$ are insect-pollinated; the rest are water-pollinated. How many are water-pollinated? [\[Reproduction in Plants\]](#)

- (A) 150
- (B) 25
- (C) 50
- (D) 20
- (E) 60
- (F) 75
- (G) 100

68. The sum of a number and its half is 27. Find the number. [\[Simple Equations\]](#)

- (A) 9
- (B) 13.5
- (C) 40.5
- (D) 36
- (E) 54
- (F) 27
- (G) 18

69. Vinegar is about 5% acetic acid by volume. A bottle holds 250 mL of vinegar. How many millilitres of it are pure acetic acid? [\[Acids, Bases & Salts\]](#)

- (A) 10 mL
- (B) 25 mL
- (C) 2.5 mL
- (D) 5 mL
- (E) 50 mL
- (F) 125 mL
- (G) 12.5 mL

70. Three-fifths of a journey is 36 km. How long is the whole journey? [\[Fractions & Decimals\]](#)

- (A) 108 km
- (B) 45 km
- (C) 30 km
- (D) 12 km
- (E) 60 km
- (F) 21.6 km

71. Pollen carried from one flower to a DIFFERENT flower of the same kind brings in more variety. What is this type of pollination called? [\[Reproduction in Plants\]](#)

- (A) Self-pollination
- (B) Fertilisation
- (C) Budding
- (D) Cross-pollination
- (E) Germination
- (F) Seed dispersal
- (G) Vegetative propagation

72. In a class the number of boys is 6 more than the number of girls, and there are 30 students in all. How many girls are there? [\[Simple Equations\]](#)

- (A) 12 girls
- (B) 18 girls
- (C) 9 girls
- (D) 15 girls
- (E) 24 girls
- (F) 6 girls
- (G) 10 girls

73. Bacteria in the mouth make acid that decays teeth, and basic toothpaste neutralises it. There is 8 mL of acid in the mouth and one brushing neutralises 6 mL. After brushing, how much acid remains and is the mouth still acidic? [\[Acids, Bases & Salts\]](#)

- (A) 14 mL, acidic
- (B) 0 mL, neutral
- (C) 6 mL, acidic
- (D) It turns colourless
- (E) 2 mL, still acidic
- (F) 2 mL, basic
- (G) 8 mL, acidic

74. Divide $\frac{5}{6}$ by $\frac{2}{3}$. Give the answer as a mixed number. [\[Fractions & Decimals\]](#)

- (A) $\frac{5}{8}$
- (B) 1
- (C) $1\frac{1}{4}$
- (D) $2\frac{1}{2}$
- (E) $1\frac{1}{2}$
- (F) $\frac{10}{9}$
- (G) $\frac{5}{9}$

75. A papaya fruit holds about 300 seeds. A farmer plants them all, and 2 out of every 5 grow into plants. How many plants grow? [\[Reproduction in Plants\]](#)
- (A) 180
 - (B) 150
 - (C) 120
 - (D) 60
 - (E) 75
 - (F) 100
 - (G) 5
76. A taxi charges Rs 50 as a fixed fee plus Rs 12 for every kilometre. A trip cost Rs 158 in total. How many kilometres was the trip? [\[Simple Equations\]](#)
- (A) 8 km
 - (B) 17 km
 - (C) 7 km
 - (D) 108 km
 - (E) 9 km
 - (F) 13 km
 - (G) 10 km
77. Fresh milk is neutral, but when it turns into curd an acid forms. What happens to blue litmus paper dipped in the curd? [\[Acids, Bases & Salts\]](#)
- (A) No change at all
 - (B) Blue litmus turns red
 - (C) It turns green
 - (D) 2 mL, neutral
 - (E) It stays blue
 - (F) Red litmus turns blue
78. Calculate $0.2 \times 0.2 \times 0.2$. [\[Fractions & Decimals\]](#)
- (A) 0.6
 - (B) 0.8
 - (C) 0.08
 - (D) 0.0008
 - (E) 0.064
 - (F) 0.008
79. Which seed is dispersed by water because it floats and has a light, fibrous coat? [\[Reproduction in Plants\]](#)
- (A) Coconut
 - (B) Cotton seed
 - (C) Balsam seed
 - (D) Pea pod seed
 - (E) Maple seed
 - (F) Cocklebur
 - (G) Dandelion seed

80. Four times a number equals that number increased by 18. Find the number. [\[Simple Equations\]](#)

- (A) 18
- (B) 24
- (C) 9
- (D) 3
- (E) 4.5
- (F) 6
- (G) 5

81. Lemon juice is acidic and baking soda is basic; mixing them fizzes until one is used up. If 3 spoons of soda neutralise 5 spoons of lemon juice, how many spoons of soda neutralise 20 spoons of lemon juice?

[\[Acids, Bases & Salts\]](#)

- (A) 8 spoons
- (B) 4 spoons
- (C) 20 spoons
- (D) 12 spoons
- (E) 10 spoons
- (F) 33 spoons
- (G) 15 spoons

82. A pizza is cut into 8 equal slices. You eat 3 slices and a friend eats 2. What fraction of the pizza is left, written as a decimal? [\[Fractions & Decimals\]](#)

- (A) 0.375
- (B) 0.25
- (C) 0.4
- (D) 0.625
- (E) 0.3
- (F) 0.5
- (G) 0.125

83. Bryophyllum grows new plants from buds on the edges of its leaves. One leaf has 8 buds. If 3 such leaves are placed in soil and 75% of all the buds grow, how many new plants form? [\[Reproduction in Plants\]](#)

- (A) 12
- (B) 20
- (C) 18
- (D) 6
- (E) 16
- (F) 21

84. Half of a number, plus 7, gives 19. Find the number. [\[Simple Equations\]](#)

- (A) 6
- (B) 52
- (C) 20
- (D) 26
- (E) 24
- (F) 16
- (G) 12

- 85.** Red cabbage juice is red in acid, purple when neutral, and green in base. A solution turns it green; then lemon juice is added slowly until it goes purple and then red. What does this show? [\[Acids, Bases & Salts\]](#)
- (A) It was a strong acid
 - (B) It was neutral all along
 - (C) It was a salt that cannot change
 - (D) The colours mean nothing about acids and bases
 - (E) Lemon juice is a base
 - (F) It was acidic all along
 - (G) It was basic; the acid neutralised it and then made it acidic
- 86.** Add the mixed numbers $2\frac{3}{4}$ and $1\frac{1}{2}$. [\[Fractions & Decimals\]](#)
- (A) $3\frac{1}{4}$
 - (B) $4\frac{1}{2}$
 - (C) $4\frac{1}{6}$
 - (D) $3\frac{4}{6}$
 - (E) $4\frac{3}{4}$
 - (F) $3\frac{1}{2}$
 - (G) $4\frac{1}{4}$
- 87.** A flower opens at 6 a.m., and its stigma stays ready to receive pollen for 9 hours. By what time must pollen arrive for pollination to succeed? [\[Reproduction in Plants\]](#)
- (A) 3 p.m.
 - (B) 3 a.m.
 - (C) 9 p.m.
 - (D) 1 p.m.
 - (E) 9 a.m.
 - (F) 12 noon
- 88.** A tank holds some litres of water. After 18 litres are added, it holds 50 litres. Form an equation and find how much it held at first. [\[Simple Equations\]](#)
- (A) 32 litres
 - (B) 16 litres
 - (C) 30 litres
 - (D) 50 litres
 - (E) 18 litres
 - (F) 68 litres
 - (G) 36 litres
- 89.** 30 mL of an acid needs 30 mL of a base to be neutralised exactly. You add only 18 mL of base. How much more base is needed, and what is the mixture right now? [\[Acids, Bases & Salts\]](#)
- (A) 12 mL more; basic
 - (B) 48 mL more; acidic
 - (C) 30 mL more; acidic
 - (D) 18 mL more; neutral
 - (E) 12 mL more; neutral
 - (F) 12 mL more; still acidic
 - (G) 0 mL more; neutral

90. A 4.5 kg bag of rice is used at 0.75 kg per day. For how many full days does the bag last? [\[Fractions & Decimals\]](#)

- (A) 4 days
- (B) 5 days
- (C) 3 days
- (D) 6 days
- (E) 8 days
- (F) 7 days
- (G) 9 days

91. Put these events in the correct order: seed forms, fruit ripens, pollination, fertilisation. [\[Reproduction in Plants\]](#)

- (A) Fertilisation, pollination, fruit ripens, seed forms
- (B) Pollination, fertilisation, seed forms, fruit ripens
- (C) Pollination, seed forms, fertilisation, fruit ripens
- (D) Fruit ripens, pollination, fertilisation, seed forms
- (E) Seed forms, pollination, fertilisation, fruit ripens
- (F) Fertilisation, pollination, seed forms, fruit ripens
- (G) Pollination, fertilisation, fruit ripens, seed forms

92. The three angles of a triangle are x , $2x$ and $3x$ degrees. Find the value of x . [\[Simple Equations\]](#)

- (A) 36 degrees
- (B) 60 degrees
- (C) 20 degrees
- (D) 90 degrees
- (E) 30 degrees
- (F) 15 degrees

93. A small tank of acidic water needs 0.25 kg of base to fix. A swimming pool that is 80 times larger needs how much base? [\[Acids, Bases & Salts\]](#)

- (A) 200 kg
- (B) 0.25 kg
- (C) 20.25 kg
- (D) 40 kg
- (E) 2 kg
- (F) 20 kg
- (G) 16 kg

94. Work out 0.144 divided by 1.2. [\[Fractions & Decimals\]](#)

- (A) 1.44
- (B) 0.2
- (C) 0.012
- (D) 12
- (E) 0.0012
- (F) 0.12
- (G) 1.2

- 95.** A mushroom releases 1,00,000 spores, but only 0.01% of them land somewhere they can grow. How many spores can grow? [\[Reproduction in Plants\]](#)
- (A) 1000
 - (B) 10
 - (C) 1
 - (D) 10000
 - (E) 0
 - (F) 100
 - (G) 50
- 96.** Five times a number is the same as three times the number plus 14. Find the number. [\[Simple Equations\]](#)
- (A) 10
 - (B) 28
 - (C) 14
 - (D) 9
 - (E) 5
 - (F) 7
- 97.** An acid and a base react in the volume ratio 4 parts acid to 5 parts base. To neutralise 1 litre (1000 mL) of the acid, how many millilitres of base are needed? [\[Acids, Bases & Salts\]](#)
- (A) 1250 mL
 - (B) 900 mL
 - (C) 500 mL
 - (D) 2000 mL
 - (E) 1125 mL
 - (F) 1000 mL
- 98.** In a class, 0.4 of the students walk to school, 0.35 cycle, and the remaining 10 students take the bus. How many students are in the class? [\[Fractions & Decimals\]](#)
- (A) 80
 - (B) 50
 - (C) 20
 - (D) 25
 - (E) 10
 - (F) 40
 - (G) 30
- 99.** A bean seed splits into two halves (cotyledons) that store food. If 5 bean seeds are split open, how many cotyledons are there in total, and what is such a two-seed-leaf plant called? [\[Reproduction in Plants\]](#)
- (A) 10 cotyledons; it is a fern
 - (B) 2 cotyledons; it is a dicot
 - (C) 10 cotyledons; it is a dicot
 - (D) 5 cotyledons; it is a monocot
 - (E) 20 cotyledons; it is a dicot
 - (F) 15 cotyledons; it is a dicot
 - (G) 10 cotyledons; it is a monocot

100. Reena is 3 years older than Meena, and the sum of their ages is 27 years. How old is Meena? [\[Simple](#)

[Equations\]](#)

- (A) 13.5 years
- (B) 15 years
- (C) 10 years
- (D) 30 years
- (E) 9 years
- (F) 24 years
- (G) 12 years